BEU CONNECT

Aryabhatta Knowledge University, Patna

B.Tech. 4th Semester Examination, 2018

Time: 03 Hours

Code: 031404 ***Subject: POWER SYSTEM-I*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option (any seven): **[14]**

(a) Which of the following distribution systems is preferred for good efficiency and high economy?
(i) Single-phase, 2-wire system
(ii) 2-phase, 3-wire system
(iii) 3-phase, 3-wire system
(iv) 3-phase, 4-wire system

(b) The main drawback(s) of overhead system over underground system is/are
(i) underground system is more flexible than overhead system
(ii) higher charging current
(iii) surge problem
(iv) high initial cost

(c) The highest transmission voltage used in India is
(i) 400 kV
(ii) 220 kV
(iii) 132 kV
(iv) 765 kV

(d) The function of steel wire in an ACSR conductor is to
(i) compensate for skin effect
(ii) take care of surges
(iii) provide additional mechanical strength
(iv) reduce inductance

(e) Corona is accompanied by
(i) violet visible discharge in darkness
(ii) hissing sound
(iii) power loss
(iv) All of the above

(f) The conductor carries more current on the surface in comparison to its core. This phenomenon is called the
(i) skin effect
(ii) corona
(iii) Ferranti effect
(iv) Lenz's effect

(g) Ferranti effect happens in transmission line when the line is
(i) short and loaded
(ii) long and loaded
(iii) long and unloaded
(iv) None of the above

(h) A transmission line is distortionless if
(i) $RG = LC$
(ii) $RC = GL$
(iii) $R/C = G/L$
(iv) $R = G$

(i) The receiving-end voltage of a transmission line will be greater than the sending-end voltage if the load is
(i) greater than SIL (surge impedance loading)
(ii) less than SIL
(iii) equal to SIL
(iv) None of the above

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

(j) Capacitor grading of cable means

(i) use of dielectrics in different concentrations

(ii) introduction of capacitances at various lengths of cable to counter the effect of inductance

(iii) use of dielectric of different permittivities

(iv) grading according to capacitance per km length of the cable

Q.2 Solve both questions : **[14]**

(a) Explain 3-phase, 4-wire system of distribution of electrical power.

(b) A 2-wire feeder ABC has a load of 100 Ampere at unity p.f. at C and 65 Ampere at p.f. 0.8 lagging at B. The impedance AB is $(0.06+j0.08)\Omega$ and that of BC is $(0.04+j0.12)\Omega$. If the voltage at the far end C is to be maintained at 230 V, determine the voltage (i) at A and (ii) at B.

Q.3 Solve both questions : **[14]**

(a) Derive an expression for the capacitance of a single-phase overhead transmission line.

(b) Find out capacitance of a single-phase line 30 km long consisting of two parallel wires each 15 mm diameter and 1.5 m apart.

Q.4 Solve both questions : **[14]**

(a) A single-phase line has an impedance of $5\angle 60^{\circ}\Omega$. It is supplying a load of 120 A, 33 kV at 0.8 p.f. lagging. Calculate its regulation.

(b) In a 3-phase, 4-wire system, the line voltage is 400 V and non-inductive loads of 10, 8 and 5 kW are connected between the three line conductors and the neutral. Calculate (i) the current in each line and (ii) the current in the neutral conductor.

Q.5 Solve both questions : **[14]**

(a) What are the sources of vibrations in a transmission line? Explain the methods used to damp out these vibrations.

(b) Obtain an expression for the sag of a transmission line supported by towers of different heights at the ends.

Q.6 Solve both questions : **[14]**

(a) Define regulation of a 3-phase a.c. transmission system and develop an expression for approximate voltage regulation.

(b) Explain surge impedance loading with respect to an overhead transmission line.

Q.7 Solve both questions : **[14]**

An overhead 3-phase transmission line delivers 5 MW at 22 kV at 0.8 lagging power factor. The resistance and reactance of each conductor are 4 Ω and 6 Ω respectively. Determine: (a) Sending-end voltage (b) Percentage regulation (c) Total line losses (d) Transmission efficiency.

Q.8 Solve both questions : **[14]**

(a) With the neat labelled diagram, show the various parts of a high-voltage single-core cable.

(b) Explain briefly the following methods of grading of cables: (i) Capacitance grading (ii) Intersheath grading.

Q.9 Solve both questions : **[14]**

(a) A 132 kV transmission line has the following data: Weight of conductor = 680 kg/km, Length of span = 260 m, Ultimate strength = 3100 kg, Safety factor = 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 m.

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

BEU CONNECT

(b) The three conductors of a 3-phase transmission line are arranged in a horizontal plane and are 3 m apart. The diameter of each conductor is 4 cm. Determine the inductance per km of each phase. Assume balanced load and R, Y, B phase sequence.

BEU CONNECT

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store